BREACHMIRROR AI

Next-Generation Cybersecurity Awareness & Teachable Moments Platform

Application Documentation · Version 2.4

1. Executive Summary

BreachMirror AI is an enterprise-grade cybersecurity awareness and real-time telemetry platform that converts security incidents into zero-blame, high-retention teachable moments — built for new hires, interns, and fast-moving engineering teams. Instead of locking accounts or filing disciplinary tickets, it intercepts unsafe actions (pasting secrets into public AI prompts, committing keys to Git, connecting unverified USB drives) at the point of risk, halts downstream exposure, and reflects the moment back as an interactive, psychologically safe lesson.

> **Why It Matters**
> New joiners and interns are disproportionately responsible for security incidents in their first 30 days. BreachMirror AI closes that gap with real-time interception, AI coaching, and gamified onboarding — without fear-based reprimand.

Core Modules

Module	Purpose
Master Dashboard	Unified overview: velocity metrics, live feed, quick actions
Threat Telemetry	Streaming feed of intercepted breach events, filterable by severity
Security Playbook & Handbook	Do's/Don'ts directives plus an interactive New Joiner Security Handbook
D3 Analytics	Threat velocity, attack-vector donut, and tenure-risk comparison charts
Dilemma Lab	Branching workplace scenarios with consequence timelines
AI Threat Sandbox	Free-text analysis of custom actions with blast-radius scoring

2. Architecture & Technology Stack

A full-stack TypeScript application: a React 19 client, an Express.js server with Vite middleware, and an optional Google Gemini 3.8 Flash integration for AI-generated coaching.

Layer	Composition
Client	React 19 · TypeScript · Vite · Tailwind CSS 4 · Lucide Icons · D3.js · Decoupled MFE Event Bus
Server	Express.js · Vite Dev Middleware · In-Memory Store · AES-256-GCM profile · Heuristic Fallback
AI Layer	@google/genai SDK (server-side only) · Zero-Blame Coaching & Blast-Radius Engine

Micro-Frontend Design

Dashboard widgets are decoupled and communicate only through a shared publish/subscribe Event Bus (e.g. BREACH_DETECTED, RULE_SELECTED, WIDGET_LAYOUT_CHANGED, THEME_CHANGED), avoiding tight coupling between telemetry, playbook, simulator, and analytics domains. Widget layout and theme preference persist via localStorage.

3. Feature Highlights

Personalized Login

- Operator authentication panel with name/call-sign entry and clearance role selector; the WELCOME button dynamically displays the entered name and disappears once logged in.
- Operator name persists across the hero banner, navbar badge, and AI-generated coaching copy for the session.

Live Telemetry & AI Sandbox

- Streaming breach feed with severity filters (CRITICAL/HIGH/ELEVATED/MONITORED) and forensic detail per event.

- AI Threat Sandbox: free-text scenario input returns a 0–100 blast-radius score, remediation, and sanctioned alternative — with a built-in heuristic fallback when no Gemini API key is set.

New Joiner Security Handbook

- Eight breach vectors covered: GenAI leaks, Git secrets, AiTM phishing, BadUSB, cloud storage exposure, supply-chain typosquatting, tailgating, and PII in chat tools.
- Four-stage interactive journey (Attack Anatomy → Prevention Rules → Code Review → Instinct Quiz) ending in a printable Certified Cyber Sentinel credential.

Playbook, Checklist & Dilemma Lab

- Five policy domains (AI Tools, Git Hygiene, MFA, Devices, Cloud Storage) each pairing a sanctioned habit with a forbidden anti-pattern and real incident precedent.
- Gamified First-30-Days checklist with XP scoring, plus a branching Dilemma Lab that scores choices against a Day 0–10 consequence timeline.

Analytics, Customization & Accessibility

- D3.js threat-timeline, attack-vector donut, and tenure-risk bar charts — each with a screen-reader table fallback.
- Fully customizable, reorderable widget layout; Dark/Light/System theming with WCAG-compliant contrast.

4. Keyboard Shortcuts & API Reference

Key	Action
1–5	Switch tabs: Dashboard, Telemetry, Playbook, Analytics, Dilemma Lab
S	Open AI Threat Sandbox
R	Inject a simulated threat
C	Open widget customizer
D	Toggle Dark / Light theme
H	Open Security Handbook
? / Esc	Shortcuts help / Close modals

Endpoint	Method	Description
/api/health	GET	Heartbeat, encryption profile, Gemini readiness
/api/telemetry/events	GET	Active & historic breach events
/api/telemetry/simulate	POST	Injects a simulated breach incident
/api/ai/teachable-moment	POST	Zero-blame AI coaching for an event
/api/ai/analyze-action	POST	Blast-radius analysis of a custom action

5. Getting Started

Command	Purpose
npm install	Install dependencies
cp .env.example .env	Configure GEMINI_API_KEY (optional — falls back to heuristic engine)
npm run dev	Full-stack dev server at http://localhost:3000
npm run build	Vite client build + esbuild server bundle
npm run start	Run the production server

Zero-Blame Philosophy

Principle	In Practice
Safety First	Threats intercepted in real time, before data exfiltration
Zero-Blame Coaching	Explains why the tool intervened and what the employee was trying to do
Instant Sanctioned Path	Surfaces the exact approved tool or workflow so work continues without delay

Design Intent

Every modal and coaching message is written to educate first and never to shame — reinforcing that reporting a near-miss is itself the desired security behavior.